Sol'n to MP363 Winter 2015-2016 exam

P.1

(a) The eq'n $\frac{\mu v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$ comes from the fact that the attractive coulomb force must be equal to the centripetal force in order for the electron to maintain its circular orbit. From this, we can immediately get an expression for v in terms of r:

$$v = \sqrt{\frac{e^2}{4\pi\epsilon_0 \mu r}}$$

and so the energy is

$$E = \frac{1}{2}\mu v^2 - \frac{e^2}{4\pi\epsilon_0 r} = \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 r}$$

and the angular momentum is

$$L = \mu v r = \mu \sqrt{\frac{e^2}{4\pi\epsilon_0 \mu r}}\, r = \sqrt{\frac{\mu e^2 r}{4\pi\epsilon_0}}.$$

Thus, we can eliminate r in favour of L:

$$r = \frac{4\pi\epsilon_0 L^2}{\mu e^2}$$

to get

$$E = -\frac{e^2}{8\pi\epsilon_0}\left(\frac{\mu e^2}{4\pi\epsilon_0 L^2}\right) = -\frac{\mu e^4}{32\pi^2\epsilon_0^2 L^2}.$$

Thus, if $L_n = n\hbar$ are the allowed quantized values of the AM, we see immediately that the energy is quantized as

$$\boxed{E_n = -\frac{\mu e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2}}$$ [10 pts]

(b) We already have r in terms of L, so putting $L_n = n\hbar$ gives

$$\boxed{r_n = \frac{4\pi\epsilon_0}{\mu e^2}L_n^2 = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2}n^2}.$$ [10 pts]

And now that we have r_n, we get v_n:

$$v_n = \sqrt{\frac{e^2}{4\pi\epsilon_0\mu r_n}} = \sqrt{\frac{e^2}{4\pi\epsilon_0\mu}\cdot\frac{\mu e^2}{4\pi\epsilon_0\hbar^2}\frac{1}{n^2}} = \boxed{\frac{e^2}{4\pi\epsilon_0\hbar}\frac{1}{n}}$$ [10 pts]

(d) The proton is nearly 2000 times more massive than the electron, so this means $m_p + m_e \approx m_p$ to a very good approximation (within about 0.2%). Therefore, the reduced mass is

$$\mu = \frac{m_e m_p}{m_e + m_p} \approx \frac{m_e m_p}{m_p} = m_e$$

and so $\boxed{\mu \text{ may be replaced by } m_e \text{ in all the above expressions}}$ [5 pts] and still get pretty accurate predictions. So let's do that for the quantities asked for:

2

$$E_1 \approx -\frac{m_e e^4}{32\pi^2\epsilon_0^2\hbar^2} = -\frac{(9.11\times10^{-27}\,kg)(1.60\times10^{-19}\,C)^4}{32\pi^2(8.85\times10^{-12}\,C^2N^{-1}m^{-1})^2(1.05\times10^{-34}\,J\cdot s)^2}$$

$$= -2.18\times10^{-18}\,J$$

$$= \boxed{-13.6\,eV}$$ [5 pts]

$$r_1 \approx \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} = \frac{4\pi(8.85\times10^{-12}\,C^2N^{-1}m^{-2})(1.05\times10^{-34}\,J\cdot s)^2}{(9.11\times10^{-31}\,kg)(1.60\times10^{-19}\,C)^2}$$

$$= 5.26\times10^{-11}\,m = \boxed{0.53\,Å}$$ [5 pts]

$$v_1 = \frac{e^2}{4\pi\epsilon_0\hbar} = \frac{(1.60\times10^{-19}\,C)^2}{4\pi(8.85\times10^{-12}\,C^2N^{-1}m^{-1})(1.05\times10^{-34}\,J\cdot s)} = 2.19\times10^6\,m\cdot s^{-1}$$

so $v_1/c = 7.3\times10^{-3}c$ = $\boxed{\text{0.73\% of the speed of light.}}$ [5 pts]

P.2

(a) "Orthonormal" has its roots in "ortho" – Greek for "right", in the sense of "right angle" – and "normal" – very normalized to a particular length, 1 in this case. A set of vectors is orthonormal if (i) two different vectors have a zero inner product with one another and (ii) each vector has unit norm. In mathematical notation,

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}$$ [5 pts]

where δ_{ij} is the Kronecker-δ symbol.

(b) Now suppose $|\psi\rangle$ is a vector that's a linear combination of the vectors in (a), i.e. $|\psi\rangle = \sum_{j=1}^{n} c_j|\phi_j\rangle$. How do we determine the coefficients? Take the inner product of this vector with the vector $\langle\phi_i|$: this gives

$$\langle\phi_i|\psi\rangle = \langle\phi_i|\sum_{j=1}^{n} c_j\phi_j\rangle = \sum_{j=1}^{n}\langle\phi_i|c_j\phi_j\rangle \quad (\text{since } \langle\psi|\chi+\xi\rangle = \langle\psi|\chi\rangle+\langle\psi|\xi\rangle)$$

$$= \sum_{j=1}^{n} c_j\langle\phi_i|\phi_j\rangle \quad (\text{since } \langle\psi|a\chi\rangle = a\langle\psi|\chi\rangle)$$

$$= \sum_{j=1}^{n} c_j\delta_{ij} \quad (\text{orthonormality}).$$

$\delta_{ij} = 0$ for all values of j except for $j=i$. Thus, the only term in the sum that is nonzero is the $j=i$ term, giving

$$\boxed{c_i = \langle\phi_i|\psi\rangle}$$ [4 pts]

And so if $|\psi\rangle$ is known, we know exactly what coefficients to use to build it out of $|\phi_1\rangle, \ldots, |\phi_n\rangle$.

(c) Since we say $\langle\psi|\xi\rangle$ is an inner product, $\langle\psi|\xi\rangle = \langle\xi|\psi\rangle^*$, which means that if $\langle\psi|\xi\rangle = 0$, then $\langle\xi|\psi\rangle$ is automatically zero as well. Thus, we only need to compute $\langle\phi_0|\phi_0\rangle$, $\langle\phi_0|\phi_1\rangle$, $\langle\phi_0|\phi_2\rangle$, $\langle\phi_1|\phi_1\rangle$, $\langle\phi_1|\phi_2\rangle$ and $\langle\phi_2|\phi_2\rangle$, because, for example, $\langle\phi_0|\phi_1\rangle = 0 \Rightarrow \langle\phi_1|\phi_0\rangle = 0$.

$\phi_0(x) = \frac{1}{\sqrt{2L}}$, $\phi_1(x) = \frac{1}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)$ and $\phi_2(x) = \frac{1}{\sqrt{L}}\cos\left(\frac{\pi x}{L}\right)$, so

$$\langle\phi_0|\phi_0\rangle = \int_{-L}^{L}\left(\frac{1}{\sqrt{2L}}\right)^*\left(\frac{1}{\sqrt{2L}}\right)dx = \frac{1}{2L}\int_{-L}^{L}dx = \frac{1}{2L}\left(x\right|_{-L}^{L} \boxed{=1}$$ [6 pts]

$$\langle\phi_1|\phi_1\rangle = \int_{-L}^{L}\left(\frac{1}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)\right)^*\left(\frac{1}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)\right)dx = \frac{1}{L}\int_{-L}^{L}\sin^2\left(\frac{\pi x}{L}\right)dx$$
$$= \frac{1}{2L}\int_{-L}^{L}\left[1-\cos\left(\frac{2\pi x}{L}\right)\right]dx = \frac{1}{2L}\left(x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right|_{-L}^{L} = \frac{1}{2L}(2L-0)$$
$$= \boxed{1}$$ [6 pts]

$$\langle\phi_2|\phi_2\rangle = \int_{-L}^{L}\left(\frac{1}{\sqrt{L}}\cos\left(\frac{\pi x}{L}\right)\right)^*\left(\frac{1}{\sqrt{L}}\cos\left(\frac{\pi x}{L}\right)\right)dx = \frac{1}{L}\int_{-L}^{L}\cos^2\left(\frac{\pi x}{L}\right)dx$$
$$= \frac{1}{2L}\int_{-L}^{L}\left[1+\cos\left(\frac{2\pi x}{L}\right)\right]dx = \frac{1}{2L}\left(x + \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right|_{-L}^{L} = \frac{1}{2L}(2L-0)$$
$$= \boxed{1}.$$ [6 pts]

so all functions have unit norm. Now for inner products of different functions:

$$\langle\phi_0|\phi_1\rangle = \int_{-L}^{L}\left(\frac{1}{\sqrt{2L}}\right)^*\left(\frac{1}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)\right)dx = \frac{1}{\sqrt{2}L}\int_{-L}^{L}\sin\left(\frac{\pi x}{L}\right)dx = \frac{1}{\sqrt{2}L}\left(-\frac{L}{\pi}\cos\left(\frac{\pi x}{L}\right)\right|_{-L}^{L}$$
$$= -\frac{1}{\sqrt{2}\pi}\left(\cos\pi - \cos(-\pi)\right) = -\frac{1}{\sqrt{2}\pi}(-1-(-1)) \boxed{=0}$$ [6 pts]

$$\langle\phi_0|\phi_2\rangle = \int_{-L}^{L}\left(\frac{1}{\sqrt{2L}}\right)^*\left(\frac{1}{\sqrt{L}}\cos\left(\frac{\pi x}{L}\right)\right)dx = \frac{1}{\sqrt{2}L}\int_{-L}^{L}\cos\left(\frac{\pi x}{L}\right)dx = \frac{1}{\sqrt{2}L}\left(\frac{L}{\pi}\sin\left(\frac{\pi x}{L}\right)\right)_{-L}^{L}$$
$$= \frac{1}{\sqrt{2}L}(0-0) \boxed{=0}$$ [6 pts]

$$\langle\phi_1|\phi_2\rangle = \int_{-L}^{L}\left(\frac{1}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)\right)^*\left(\frac{1}{\sqrt{L}}\cos\left(\frac{\pi x}{L}\right)\right)dx = \frac{1}{L}\int_{-L}^{L}\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)dx$$
$$= \frac{1}{2L}\int_{-L}^{L}\sin\left(\frac{2\pi x}{L}\right)dx = \frac{1}{2L}\left(-\frac{L}{2\pi}\cos\left(\frac{2\pi x}{L}\right)\right) = -\frac{1}{4\pi}\left[\cos(2\pi)-\cos(-2\pi)\right)$$
$$= -\frac{1}{4\pi}(1-1) \boxed{=0}$$ [6 pts]

$\boxed{\text{So } \langle\phi_i|\phi_j\rangle = 0 \text{ if } i\neq j \text{ and } 1 \text{ if } i=j\text{, which means the functions form an orthonormal set.}}$

P.3

(a) $|\psi\rangle$ has the form $|\psi\rangle = \sum_{n=1}^{\infty} c_n|n\rangle$ with $c_0 = \frac{1}{2}$, $c_1 = -\frac{\sqrt{3}}{2}$ and all other coefficients zero. The probability of getting $E_n = \hbar\omega(n+\frac{1}{2})$ upon measurement of the energy is $|c_n|^2$, so there is no chance of getting E_n for $n=2$ or higher, since all these coefficients are zero. Therefore, $\boxed{E_0 = \frac{1}{2}\hbar\omega \text{ and } E_1 = \frac{1}{2}\hbar\omega}$ are the possible results of an energy measurement. [5 pts each]

$|c_0|^2 = \left|\frac{i}{2}\right|^2 = \frac{1}{4}$ and $|c_1|^2 = \left|-\frac{\sqrt{3}}{2}\right|^2 = \frac{3}{4}$, so the chance of getting $\frac{1}{2}\hbar\omega$ is $\boxed{25\%}$ and the chance of getting $\frac{3}{2}\hbar\omega$ is $\boxed{75\%}$. [5 pts each]

(b) Because the state is written as a linear combination of the energy eigenstates, the expectation value of H is $\langle H\rangle = \sum_{n=0}^{\infty} |c_n|^2 E_n$. Only the first two coefficients are nonzero, so

$$\langle H\rangle = |c_0|^2 E_0 + |c_1|^2 E_1 = \left|\frac{i}{2}\right|^2 \frac{1}{2}\hbar\omega + \left|\frac{\sqrt{3}}{2}\right|^2 \frac{3}{2}\hbar\omega$$

$$= \boxed{\frac{5}{4}\hbar\omega}$$ [10 pts]

(c) Using the hint, we find that since $a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(X - \frac{iP}{m\omega}\right)$ and $a = \sqrt{\frac{m\omega}{2\hbar}}\left(X + \frac{iP}{m\omega}\right)$ gives $a^\dagger + a = 2\sqrt{\frac{m\omega}{2\hbar}}X$ and $a^\dagger - a = \frac{-2i}{m\omega}\sqrt{\frac{m\omega}{2\hbar}}P$, and so

$$X = \sqrt{\frac{\hbar}{2m\omega}}(a^\dagger + a), \quad P = i\sqrt{\frac{m\hbar\omega}{2}}(a^\dagger - a)$$

The reason this is a convenient way to write X and P is because we know exactly how $a^\dagger$ and a act on $|0\rangle$ and $|1\rangle$:

$$a^\dagger|0\rangle = \sqrt{0+1}\,|0+1\rangle = |1\rangle, \quad a^\dagger|1\rangle = \sqrt{1+1}\,|1+1\rangle = \sqrt{2}\,|2\rangle$$

$$a|0\rangle = \sqrt{0}\,|0-1\rangle = 0, \quad a|1\rangle = \sqrt{1}\,|1-1\rangle = |0\rangle$$

so

$$a^\dagger|\psi\rangle = \frac{i}{2}a^\dagger|0\rangle - \frac{\sqrt{3}}{2}a^\dagger|1\rangle = \frac{i}{2}|1\rangle - \sqrt{\frac{3}{2}}|2\rangle$$

$$a|\psi\rangle = \frac{i}{2}a|0\rangle - \frac{\sqrt{3}}{2}a|1\rangle = -\frac{\sqrt{3}}{2}|0\rangle.$$

Therefore,

$$X|\psi\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(a^\dagger|\psi\rangle + a|\psi\rangle\right) = \sqrt{\frac{\hbar}{2m\omega}}\left(-\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle - \sqrt{\frac{3}{2}}|2\rangle\right)$$

$$P|\psi\rangle = i\sqrt{\frac{m\hbar\omega}{2}}\left(a^\dagger|\psi\rangle - a|\psi\rangle\right) = i\sqrt{\frac{m\hbar\omega}{2}}\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle - \sqrt{\frac{3}{2}}|2\rangle\right)$$

The expectation values are therefore

$$\langle X\rangle = \langle\psi|X|\psi\rangle = \left(-\frac{i}{2}\langle 0| - \frac{\sqrt{3}}{2}\langle 1|\right)\left[\sqrt{\frac{\hbar}{2m\omega}}\left(-\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle - \sqrt{\frac{3}{2}}|2\rangle\right)\right]$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{i\sqrt{3}}{4}\langle 0|0\rangle - \frac{i\sqrt{3}}{4}\langle 1|1\rangle\right) \boxed{= 0}$$ [10 pts]

(since $\langle 0|1\rangle = \langle 0|2\rangle = \langle 1|0\rangle = \langle 1|2\rangle = 0$) Similarly,

$$\langle P\rangle = \langle\psi|P|\psi\rangle = \left(-\frac{i}{2}\langle 0| - \frac{\sqrt{3}}{2}\langle 1|\right)\left[i\sqrt{\frac{m\hbar\omega}{2}}\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle - \sqrt{\frac{3}{2}}|2\rangle\right)\right]$$

$$= i\sqrt{\frac{m\hbar\omega}{2}}\left(-\frac{i\sqrt{3}}{4}\langle 0|0\rangle - \frac{i\sqrt{3}}{4}\langle 1|1\rangle\right)$$

$$= \boxed{\sqrt{\frac{3m\hbar\omega}{8}}}$$ [10 pts]

so if we measured the position and momentum of this state, on average it would be just as often to the right of 0 as to the left, but its general motion would be to the right with an average value of $\sqrt{3m\hbar\omega/8}$.